

CSE 219 Practice Sheet

FFT, Sampling, Laplace Transform, and z-Transform

Contents

1	Radix-2 DIT Cooley-Tukey FFT	1
2	2D Matrix Formulation of Cooley-Tukey (Bailey’s FFT)	3
3	Bluestein’s FFT	5
4	Application of FFT: Fast Convolution	7
5	Ideal Reconstruction, Zero-Order Hold, Linear Interpolation	10
6	Laplace Transform: Definition, Properties, ROC, Poles and Zeros	12
7	Inverse Laplace Transform	14
8	Applications of Laplace Transform	17
9	z-Transform: Definition, Properties, ROC, Poles and Zeros	19
10	Inverse z-Transform	22
11	Applications of z-Transform: Causality, Stability from ROC	24

1. Radix-2 DIT Cooley-Tukey FFT

Problem 1.1

Question. Starting from $X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn}$, derive the radix-2 DIT split into even and odd subsequences and the butterfly relations for $0 \leq k < N/2$.

Solution. Splitting $n = 2r$ and $n = 2r + 1$,

$$X[k] = \sum_{r=0}^{N/2-1} x[2r]W_{N/2}^{kr} + W_N^k \sum_{r=0}^{N/2-1} x[2r + 1]W_{N/2}^{kr} = G[k] + W_N^k H[k].$$

Using $W_N^{k+N/2} = -W_N^k$,

$$X[k + N/2] = G[k] - W_N^k H[k], \quad 0 \leq k < \frac{N}{2}.$$

```
graph LR; x[n] --> split[even / odd split]; split --> G[k]; split --> H[k]; G[k] --> top(( )); H[k] --> top; G[k] --> bottom(( )); H[k] --> bottom; top --> X[k]; bottom --> X[k + N/2];
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Problem 1.2

Question. Compute the 4-point DIT FFT of $x[n] = [2, 1, -3, 4]$ using radix-2 butterflies.

Solution.

$$x_e = [2, -3], \quad x_o = [1, 4].$$

The two 2-point DFTs are

$$G = [-1, 5], \quad H = [5, -3].$$

Now combine with

$$X[k] = G[k] + W_4^k H[k], \quad X[k+2] = G[k] - W_4^k H[k], \quad k = 0, 1.$$

For $k = 0$,

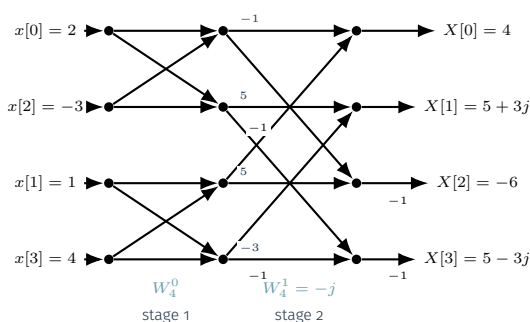
$$X[0] = -1 + 5 = 4, \quad X[2] = -1 - 5 = -6.$$

For $k = 1$, $W_4^1 = -j$, so

$$X[1] = 5 + (-j)(-3) = 5 + 3j, \quad X[3] = 5 - (-j)(-3) = 5 - 3j.$$

Hence

$$X[k] = [4, 5 + 3j, -6, 5 - 3j].$$

**Problem 1.3**

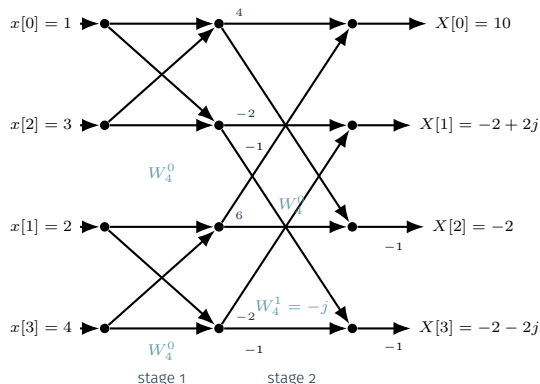
Question. Compute the 4-point FFT of $x[n] = [1, 2, 3, 4]$ using two 2-point butterflies rather than the direct DFT sum.

Solution. Split into even and odd subsequences:

$$x_e = [1, 3], \quad x_o = [2, 4].$$

Their 2-point DFTs are $G = [4, -2]$ and $H = [6, -2]$. Therefore

$$X[0] = 10, \quad X[1] = -2 + 2j, \quad X[2] = -2, \quad X[3] = -2 - 2j.$$

**Problem 1.4**

Question. Compute the 8-point FFT of $x[n] = [1, 2, 0, -1, 3, 0, 2, 1]$ using the full radix-2 DIT butterfly. Show the stage values, not only the final merge.

Solution. For an iterative radix-2 DIT flow, place the data in bit-reversed order:

$$[x[0], x[4], x[2], x[6], x[1], x[5], x[3], x[7]] = [1, 3, 0, 2, 2, 0, -1, 1].$$

After the first butterfly stage,

$$[4, -2, 2, -2, 2, 2, 0, -2].$$

After the second stage,

$$[6, -2 + 2j, 2, -2 - 2j, 2, 2 + 2j, 2, 2 - 2j].$$

Now use the final twiddle factors W_8^k , $k = 0, 1, 2, 3$:

$$X[0] = 6 + W_8^0(2) = 8,$$

$$X[1] = (-2 + 2j) + W_8^1(2) = (-2 + 2j) + \sqrt{2}(1 - j) = -2 + 2\sqrt{2} + (2 - \sqrt{2})j,$$

$$X[2] = 2 + W_8^2(2) = 2 - 2j,$$

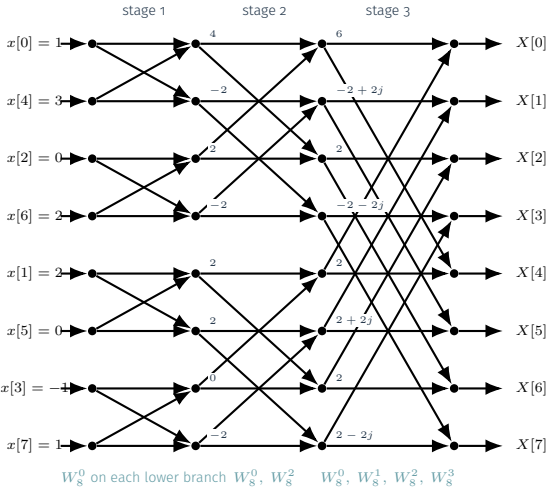
$$X[3] = (-2 - 2j) + W_8^3(2) = -2 - 2\sqrt{2} - (2 + \sqrt{2})j,$$

and

$$\begin{aligned} X[4] &= 6 - W_8^0(2) = 4, \\ X[5] &= (-2 + 2j) - W_8^1(2) = -2 - 2\sqrt{2} + (2 + \sqrt{2})j, \\ X[6] &= 2 - W_8^2(2) = 2 + 2j, \\ X[7] &= (-2 - 2j) - W_8^3(2) = -2 + 2\sqrt{2} + (\sqrt{2} - 2)j. \end{aligned}$$

So the final 8-point FFT is

$$X[k] = [8, -2 + 2\sqrt{2} + (2 - \sqrt{2})j, 2 - 2j, -2 - 2\sqrt{2} - (2 + \sqrt{2})j, 4, -2 - 2\sqrt{2} + (2 + \sqrt{2})j, 2 + 2j, -2 + 2\sqrt{2} + (\sqrt{2} - 2)j].$$



Problem 1.5

Question. Recover $x[n]$ from

$$X[k] = [6, \sqrt{2} - j\sqrt{2}, 2 + 4j, -\sqrt{2} - j\sqrt{2}, -2, -\sqrt{2} + j\sqrt{2}, 2 - 4j, \sqrt{2} + j\sqrt{2}]$$

using an 8-point inverse FFT.

Solution. Use the same radix-2 DIT butterfly graph as the forward 8-point FFT, but replace each lower-branch twiddle by W_8^{-k} and divide the final output by 8. Feed the samples in bit-reversed order:

$$[X[0], X[4], X[2], X[6], X[1], X[5], X[3], X[7]] = [6, -2, 2 + 4j, 2 - 4j, \sqrt{2} - j\sqrt{2}, -\sqrt{2} + j\sqrt{2}, -\sqrt{2} - j\sqrt{2}, \sqrt{2} + j\sqrt{2}].$$

Stage 1 uses only $W_8^0 = 1$, so

$$\text{stage 1: } [4, 8, 4, 8j, 0, 2\sqrt{2} - 2\sqrt{2}j, 0, -2\sqrt{2} - 2\sqrt{2}j].$$

At stage 2 the nontrivial lower-branch twiddle is $W_8^{-2} = j$, giving

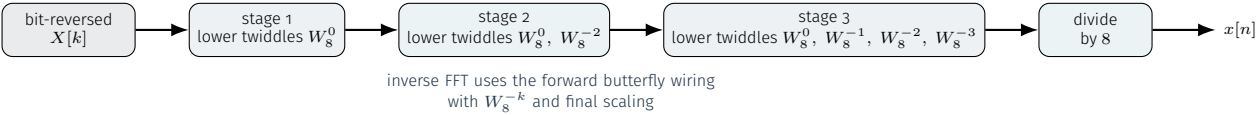
$$\text{stage 2: } [8, 0, 0, 16, 0, 4\sqrt{2} - 4\sqrt{2}j, 0, 0].$$

At stage 3 the lower-branch twiddles are $W_8^0, W_8^{-1}, W_8^{-2}, W_8^{-3}$, so

$$\text{stage 3: } [8, 8, 0, 16, 8, -8, 0, 16].$$

Therefore

$$x[n] = \frac{1}{8}[8, 8, 0, 16, 8, -8, 0, 16] = [1, 1, 0, 2, 1, -1, 0, 2].$$



2. 2D Matrix Formulation of Cooley-Tukey (Bailey's FFT)

Problem 2.1

Question. Show how a 4-point DFT can be written as a 2×2 matrix factorization using column-major input, row DFTs, twiddles, and column DFTs.

Solution. Arrange the samples in *column-major* order so the first row contains the even-indexed samples and the second row contains the odd-indexed samples:

$$\begin{bmatrix} x_0 & x_2 \\ x_1 & x_3 \end{bmatrix}.$$

Apply 2-point DFTs along the rows:

$$\begin{bmatrix} x_0 + x_2 & x_0 - x_2 \\ x_1 + x_3 & x_1 - x_3 \end{bmatrix}.$$

Twiddle the second row by $[1, W_4^1]$, so only the lower-right entry picks up $W_4^1 = -j$. The matrix after twiddle multiplication is

$$\begin{bmatrix} x_0 + x_2 & x_0 - x_2 \\ x_1 + x_3 & -j(x_1 - x_3) \end{bmatrix}.$$

Now apply 2-point DFTs down the columns. The column-wise 2-point FFT produces

$$\begin{bmatrix} (x_0 + x_2) + (x_1 + x_3) & (x_0 - x_2) - j(x_1 - x_3) \\ (x_0 + x_2) - (x_1 + x_3) & (x_0 - x_2) + j(x_1 - x_3) \end{bmatrix}.$$

Reading out this final matrix in row-major order gives

$$\begin{aligned} X[0] &= x_0 + x_1 + x_2 + x_3, \\ X[1] &= (x_0 - x_2) - j(x_1 - x_3), \\ X[2] &= (x_0 + x_2) - (x_1 + x_3), \\ X[3] &= (x_0 - x_2) + j(x_1 - x_3). \end{aligned}$$



Problem 2.2

Question. Use a 2×4 Bailey factorization to compute the 8-point DFT of $x[n] = [1, 3, -1, 2, 0, 4, 2, 1]$.

Solution. Write the input in column-major order so the top row is the even-indexed subsequence and the bottom row is the odd-indexed subsequence:

$$\mathbf{X}_{\text{in}} = \begin{bmatrix} 1 & -1 & 0 & 2 \\ 3 & 2 & 4 & 1 \end{bmatrix}.$$

First apply a 4-point DFT to each row:

$$\mathbf{R} = \begin{bmatrix} 2 & 1+3j & 0 & 1-3j \\ 10 & -1-j & 4 & -1+j \end{bmatrix}.$$

Now twiddle the second row by $[1, W_8^1, W_8^2, W_8^3]$:

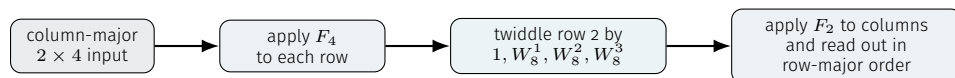
$$\mathbf{T} = \begin{bmatrix} 2 & 1+3j & 0 & 1-3j \\ 10 & (-1-j)W_8^1 & 4W_8^2 & (-1+j)W_8^3 \end{bmatrix} = \begin{bmatrix} 2 & 1+3j & 0 & 1-3j \\ 10 & -\sqrt{2} & -4j & \sqrt{2} \end{bmatrix}.$$

Finally apply 2-point DFTs down each column:

$$\begin{aligned} X[0] &= 2 + 10 = 12, & X[4] &= 2 - 10 = -8, \\ X[1] &= (1 + 3j) - \sqrt{2} = 1 - \sqrt{2} + 3j, & X[5] &= (1 + 3j) + \sqrt{2} = 1 + \sqrt{2} + 3j, \\ X[2] &= 0 - 4j = -4j, & X[6] &= 0 - (-4j) = 4j, \\ X[3] &= (1 - 3j) + \sqrt{2} = 1 + \sqrt{2} - 3j, & X[7] &= (1 - 3j) - \sqrt{2} = 1 - \sqrt{2} - 3j. \end{aligned}$$

The final matrix is read out in *row-major* order, so we obtain

$$\begin{aligned} X[0] &= 12, & X[1] &= 1 - \sqrt{2} + 3j, & X[2] &= -4j, & X[3] &= 1 + \sqrt{2} - 3j, \\ X[4] &= -8, & X[5] &= 1 + \sqrt{2} + 3j, & X[6] &= 4j, & X[7] &= 1 - \sqrt{2} - 3j. \end{aligned}$$



Problem 2.4

Question. A 16-point sequence is $x[n] = [1, 0, 0, 0, 2, 0, 0, 0, -1, 0, 0, 0, 3, 0, 0, 0]$. Use the 4×4 formulation to reduce the work.

Solution. Write the sequence in column-major form so row r collects samples with indices $n \equiv r \pmod{4}$:

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Only the first row is active. The row-stage 4-point DFT therefore acts only on

$$[1, 2, -1, 3],$$

giving

$$[5, 2 + j, -5, 2 - j].$$

So after the row DFTs,

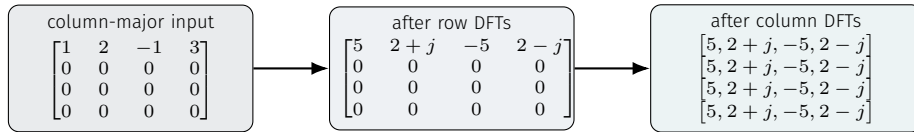
$$\mathbf{R} = \begin{bmatrix} 5 & 2+j & -5 & 2-j \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The twiddle matrix does nothing here, because only row $r = 0$ survives and $W_{16}^{rk} = W_{16}^0 = 1$. Now each column has the form $[a, 0, 0, 0]^T$, whose 4-point DFT is simply $[a, a, a, a]^T$. Hence

$$\mathbf{C} = \begin{bmatrix} 5 & 2+j & -5 & 2-j \\ 5 & 2+j & -5 & 2-j \\ 5 & 2+j & -5 & 2-j \\ 5 & 2+j & -5 & 2-j \end{bmatrix}.$$

Read out the matrix in row-major order,

$$X[k] = [5, 2+j, -5, 2-j, 5, 2+j, -5, 2-j, 5, 2+j, -5, 2-j, 5, 2+j, -5, 2-j].$$



3. Bluestein's FFT

Problem 3.1

Question. Derive Bluestein's chirp identity

$$W_N^{nk} = W_{2N}^{-(k-n)^2} W_{2N}^{k^2} W_{2N}^{n^2}.$$

Solution. Since

$$(k-n)^2 = k^2 - 2kn + n^2,$$

we get

$$kn = -\frac{(k-n)^2}{2} + \frac{k^2}{2} + \frac{n^2}{2}.$$

Because $W_N^{nk} = W_{2N}^{2nk}$, substitution gives

$$X[k] = \sum_{n=0}^{N-1} x[n] W_{2N}^{-(k-n)^2} W_{2N}^{k^2} W_{2N}^{n^2} = W_{2N}^{k^2} \sum_{n=0}^{N-1} a[n] b[k-n],$$

where $a[n] = x[n] W_{2N}^{n^2}$ and $b[m] = W_{2N}^{-m^2}$. Thus the DFT becomes a chirp-weighted linear convolution. For the needed outputs $0 \leq k \leq N-1$ and data indices $0 \leq n \leq N-1$, the difference

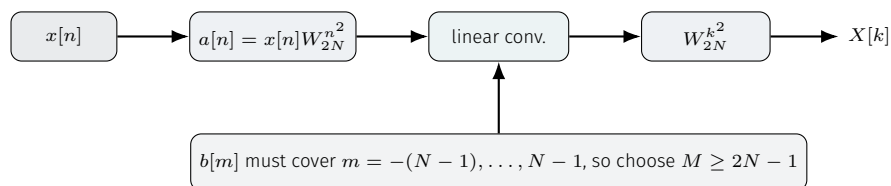
$$m = k - n$$

runs from $-(N-1)$ to $N-1$. So $b[m]$ must be available on exactly $2N-1$ consecutive indices. That is why Bluestein needs an FFT length

$$M \geq 2N-1,$$

usually the next radix-2 choice. In a packed FFT buffer, $a[n]$ sits in slots $0, \dots, N-1$, while $b[m]$ is stored as

$$[b[0], b[1], \dots, b[N-1], 0, \dots, 0, b[-(N-1)], \dots, b[-1]].$$



Problem 3.2

Question. Use Bluestein's construction to compute the 3-point DFT of $x[n] = [2, -1, 3]$.

Solution. Let $W_6 = e^{-j\pi/3}$. Bluestein writes

$$X[k] = W_6^{k^2} c[k], \quad c[k] = \sum_{n=0}^2 a[n] b[k-n],$$

with

$$a[n] = x[n] W_6^{n^2} = [2, -W_6, 3W_6^4] = \left[2, -\frac{1}{2} + \frac{\sqrt{3}}{2}j, -\frac{3}{2} + \frac{3\sqrt{3}}{2}j \right].$$

Also,

$$b[m] = W_6^{-m^2}, \quad m = -2, -1, 0, 1, 2,$$

so

$$[b[-2], b[-1], b[0], b[1], b[2]] = [W_6^{-4}, W_6^{-1}, 1, W_6^{-1}, W_6^{-4}].$$

Because $2N-1 = 5$, a radix-2 implementation uses $M = 8$ and may pack

$$A = [2, -W_6, 3W_6^4, 0, 0, 0, 0, 0],$$

$$B = [1, W_6^{-1}, W_6^{-4}, 0, 0, 0, W_6^{-4}, W_6^{-1}].$$

Now evaluate the needed convolution samples:

$$c[0] = 2 \cdot 1 + (-W_6)W_6^{-1} + 3W_6^4W_6^{-4} = 2 - 1 + 3 = 4,$$

$$c[1] = 2W_6^{-1} + (-W_6) \cdot 1 + 3W_6^4W_6^{-1} = 2W_6^{-1} - W_6 + 3W_6^3 = -\frac{5}{2} + \frac{3\sqrt{3}}{2}j,$$

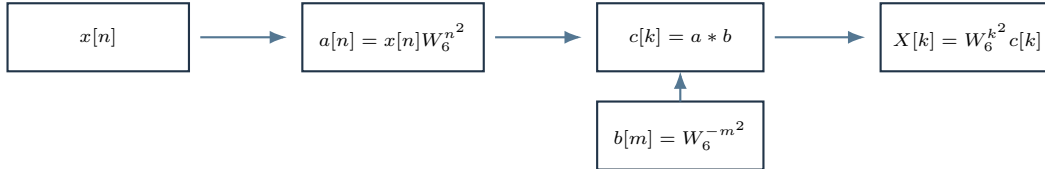
$$c[2] = 2W_6^{-4} + (-W_6)W_6^{-1} + 3W_6^4 \cdot 1 = 2W_6^2 - 1 + 3W_6^4 = -\frac{7}{2} + \frac{\sqrt{3}}{2}j.$$

Finally,

$$X[0] = W_6^0 c[0] = 4,$$

$$X[1] = W_6 c[1] = \left(\frac{1}{2} - \frac{\sqrt{3}}{2}j\right) \left(-\frac{5}{2} + \frac{3\sqrt{3}}{2}j\right) = 1 + 2\sqrt{3}j,$$

$$X[2] = W_6^4 c[2] = \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}j\right) \left(-\frac{7}{2} + \frac{\sqrt{3}}{2}j\right) = 1 - 2\sqrt{3}j.$$



Problem 3.3

Question. Run the same method on the 4-point sequence $x[n] = [1, 2, 0, 0]$ and compare with the ordinary 4-point DFT.

Solution. Let $W_8 = e^{-j\pi/4}$. Then

$$X[k] = W_8^{k^2} c[k], \quad a[n] = x[n]W_8^{n^2} = [1, 2W_8, 0, 0].$$

The chirp is $b[m] = W_8^{-m^2}$, so for $m = -3, \dots, 3$,

$$[b[-3], b[-2], b[-1], b[0], b[1], b[2], b[3]] = [W_8^{-1}, -1, W_8^{-1}, 1, W_8^{-1}, -1, W_8^{-1}].$$

Here $2N - 1 = 7$, so the usual radix-2 choice is $M = 8$, with packed buffers

$$A = [1, 2W_8, 0, 0, 0, 0, 0, 0],$$

$$B = [1, W_8^{-1}, -1, W_8^{-1}, 0, W_8^{-1}, -1, W_8^{-1}].$$

Only $a[0]$ and $a[1]$ are nonzero, hence

$$c[k] = a[0]b[k] + a[1]b[k-1], \quad 0 \leq k \leq 3.$$

Therefore

$$c[0] = 1 \cdot 1 + 2W_8 \cdot W_8^{-1} = 3,$$

$$c[1] = W_8^{-1} + 2W_8 = \frac{3\sqrt{2}}{2} - \frac{\sqrt{2}}{2}j,$$

$$c[2] = -1 + 2W_8W_8^{-1} = 1,$$

$$c[3] = W_8^{-1} - 2W_8 = -\frac{\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}j.$$

Multiplying by the output chirp gives

$$X[0] = c[0] = 3,$$

$$X[1] = W_8 c[1] = 1 - 2j,$$

$$X[2] = W_8^4 c[2] = -1,$$

$$X[3] = W_8^9 c[3] = W_8 c[3] = 1 + 2j.$$

The ordinary 4-point DFT agrees term by term:

$$X[0] = 1 + 2 = 3,$$

$$X[1] = 1 + 2W_4 = 1 - 2j,$$

$$X[2] = 1 + 2W_4^2 = -1,$$

$$X[3] = 1 + 2W_4^3 = 1 + 2j.$$

4. Application of FFT: Fast Convolution

Problem 4.1

Question. Compute the linear convolution of $x[n] = [1, 2]$ and $h[n] = [1, -1]$ using a 4-point FFT, but carry out the DFTs as radix-2 combinations of 2-point butterflies.

Solution. The output length is $2 + 2 - 1 = 3$, so a 4-point FFT is sufficient. Zero-pad both signals to length 4:

$$x_p[n] = [1, 2, 0, 0], \quad h_p[n] = [1, -1, 0, 0].$$

For $x_p[n]$,

$$G_x = \text{FFT}_2([1, 0]) = [1, 1], \quad H_x = \text{FFT}_2([2, 0]) = [2, 2].$$

So the 4-point FFT is

$$X[k] = [3, 1 - 2j, -1, 1 + 2j].$$

For $h_p[n]$,

$$G_h = \text{FFT}_2([1, 0]) = [1, 1], \quad H_h = \text{FFT}_2([-1, 0]) = [-1, -1],$$

so

$$H[k] = [0, 1 + j, 2, 1 - j].$$

Pointwise multiplication gives

$$Y[k] = X[k]H[k] = [0, 3 - j, -2, 3 + j].$$

For the inverse 4-point FFT, use the same butterfly wiring with W_4^{-k} , then divide by 4. In bit-reversed order,

$$[Y[0], Y[2], Y[1], Y[3]] = [0, -2, 3 - j, 3 + j].$$

The inverse stages are

stage 1: $[-2, 2, 6, -2j]$, stage 2: $[4, 4, -8, 0]$.

Therefore

$$y_p[n] = \frac{1}{4}[4, 4, -8, 0] = [1, 1, -2, 0].$$

Discard the padded zero and keep the linear-convolution part:

$$y[n] = [1, 1, -2].$$

Problem 4.2

Question. Compute the linear convolution of $x[n] = [1, -1, 2]$ and $h[n] = [2, 1]$ using a 4-point radix-2 FFT and an explicit inverse FFT.

Solution. Here $L = 3 + 2 - 1 = 4$, so $N = 4$ is enough. After zero-padding,

$$x_p[n] = [1, -1, 2, 0], \quad h_p[n] = [2, 1, 0, 0].$$

For $x_p[n]$,

$$G_x = \text{FFT}_2([1, 2]) = [3, -1], \quad H_x = \text{FFT}_2([-1, 0]) = [-1, -1].$$

Hence

$$X[k] = [2, -1 + j, 4, -1 - j].$$

For $h_p[n]$,

$$G_h = \text{FFT}_2([2, 0]) = [2, 2], \quad H_h = \text{FFT}_2([1, 0]) = [1, 1],$$

so

$$H[k] = [3, 2 - j, 1, 2 + j].$$

Multiply pointwise:

$$Y[k] = [6, -1 + 3j, 4, -1 - 3j].$$

For the inverse 4-point FFT, feed the spectrum in bit-reversed order and use $W_4^{-1} = j$:

$$[Y[0], Y[2], Y[1], Y[3]] = [6, 4, -1 + 3j, -1 - 3j].$$

The inverse stages are

stage 1: $[10, 2, -2, 6j]$, stage 2: $[8, -4, 12, 8]$.

Therefore

$$y[n] = \frac{1}{4}[8, -4, 12, 8] = [2, -1, 3, 2].$$

Problem 4.3

Question. Compute the linear convolution of $x[n] = [1, 0, 1]$ and $h[n] = [1, -1, 2]$. Use the smallest radix-2 FFT size and show the butterfly-based DFT steps.

Solution. The linear-convolution length is $3 + 3 - 1 = 5$, so the smallest radix-2 choice is $N = 8$. Zero-pad to length 8:

$$x_p[n] = [1, 0, 1, 0, 0, 0, 0, 0], \quad h_p[n] = [1, -1, 2, 0, 0, 0, 0, 0].$$

For $x_p[n]$, the even and odd 4-point inputs are

$$x_e = [1, 1, 0, 0], \quad x_o = [0, 0, 0, 0].$$

Their 4-point DFTs are

$$G_x = [2, 1 - j, 0, 1 + j], \quad H_x = [0, 0, 0, 0].$$

So

$$X[k] = [2, 1 - j, 0, 1 + j, 2, 1 - j, 0, 1 + j].$$

For $h_p[n]$,

$$h_e = [1, 2, 0, 0], \quad h_o = [-1, 0, 0, 0].$$

Thus

$$G_h = [3, 1 - 2j, -1, 1 + 2j], \quad H_h = [-1, -1, -1, -1].$$

Combining the 4-point branches with W_8^k gives

$$\begin{aligned} H[0] &= 3 - 1 = 2, \\ H[1] &= (1 - 2j) - W_8 = 1 - \frac{1}{\sqrt{2}} + \left(-2 + \frac{1}{\sqrt{2}}\right)j, \\ H[2] &= -1 + j, \\ H[3] &= (1 + 2j) - W_8^3 = 1 + \frac{1}{\sqrt{2}} + \left(2 + \frac{1}{\sqrt{2}}\right)j, \end{aligned}$$

and

$$\begin{aligned} H[4] &= 3 + 1 = 4, \\ H[5] &= (1 - 2j) + W_8 = 1 + \frac{1}{\sqrt{2}} + \left(-2 - \frac{1}{\sqrt{2}}\right)j, \\ H[6] &= -1 - j, \\ H[7] &= (1 + 2j) + W_8^3 = 1 - \frac{1}{\sqrt{2}} + \left(2 - \frac{1}{\sqrt{2}}\right)j. \end{aligned}$$

Now multiply pointwise:

$$\begin{aligned} Y[0] &= 4, \\ Y[1] &= (1 - j)H[1] = -1 + (-3 + \sqrt{2})j, \\ Y[2] &= 0, \\ Y[3] &= (1 + j)H[3] = -1 + (3 + \sqrt{2})j, \\ Y[4] &= 8, \\ Y[5] &= (1 - j)H[5] = -1 - (3 + \sqrt{2})j, \\ Y[6] &= 0, \\ Y[7] &= (1 + j)H[7] = -1 + (3 - \sqrt{2})j. \end{aligned}$$

For the inverse 8-point FFT, use the same radix-2 butterfly structure with W_8^{-k} . In bit-reversed order,

$$[Y[0], Y[4], Y[2], Y[6], Y[1], Y[5], Y[3], Y[7]] = [4, 8, 0, 0, -1 - (3 - \sqrt{2})j, -1 - (3 + \sqrt{2})j, -1 + (3 + \sqrt{2})j, -1 + (3 - \sqrt{2})j].$$

The inverse stages are

$$\begin{aligned} \text{stage 1: } & [12, -4, 0, 0, -2 - 6j, 2\sqrt{2}j, -2 + 6j, 2\sqrt{2}j], \\ \text{stage 2: } & [12, -4, 12, -4, -4, -2\sqrt{2} + 2\sqrt{2}j, -12j, 2\sqrt{2} + 2\sqrt{2}j], \\ \text{stage 3: } & [8, -8, 24, -8, 16, 0, 0, 0]. \end{aligned}$$

Hence

$$y_p[n] = \frac{1}{8}[8, -8, 24, -8, 16, 0, 0, 0] = [1, -1, 3, -1, 2, 0, 0, 0].$$

Therefore the linear convolution is

$$y[n] = [1, -1, 3, -1, 2].$$

Problem 4.4

Question. Repeat Problem 4.3 incorrectly with a 4-point FFT and show the circular-convolution wrap-around.

Solution. Using $N = 4$ means

$$x_p[n] = [1, 0, 1, 0], \quad h_p[n] = [1, -1, 2, 0].$$

The 4-point FFT of $x_p[n]$ is especially simple:

$$X[k] = [2, 0, 2, 0].$$

For $h_p[n]$,

$$G_h = \text{FFT}_2([1, 2]) = [3, -1], \quad H_h = \text{FFT}_2([-1, 0]) = [-1, -1],$$

so

$$H[k] = [2, -1 + j, 4, -1 - j].$$

Pointwise multiplication gives the 4-point circular spectrum

$$Y[k] = [4, 0, 8, 0].$$

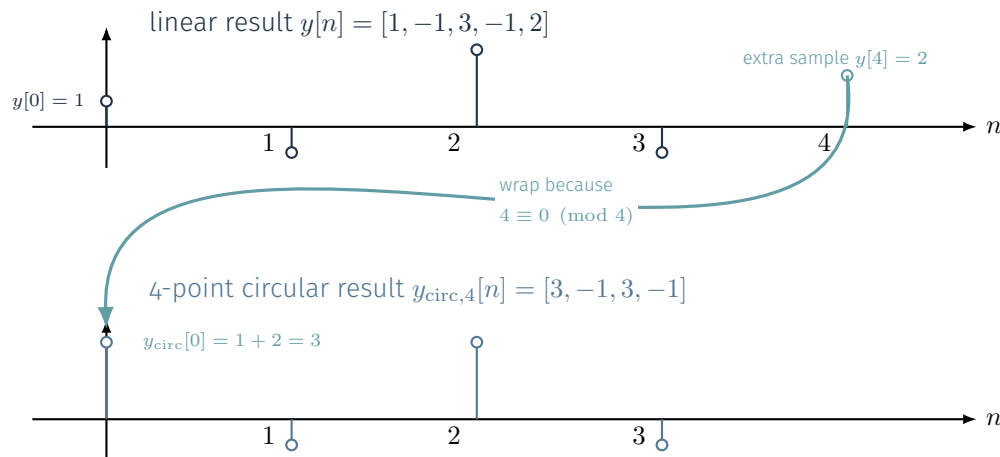
The inverse FFT of this spectrum is

$$y_{\text{circ},4}[n] = [3, -1, 3, -1].$$

Comparing with the correct linear result

$$y[n] = [1, -1, 3, -1, 2],$$

we see that the last sample 2 has wrapped and added onto the first sample 1, producing 3.



Problem 5.1

Question. State the Nyquist-Shannon sampling theorem in both angular-frequency and ordinary-frequency form.

Solution. If a continuous-time signal is bandlimited to $|\omega| \leq \omega_M$, then exact reconstruction from uniform samples is possible whenever

$$\omega_s > 2\omega_M.$$

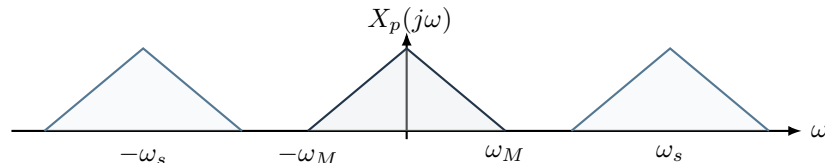
In ordinary frequency,

$$f_s > 2f_M.$$

The sampled spectrum is

$$X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s)),$$

so the theorem is really a non-overlap condition on these repeated copies.

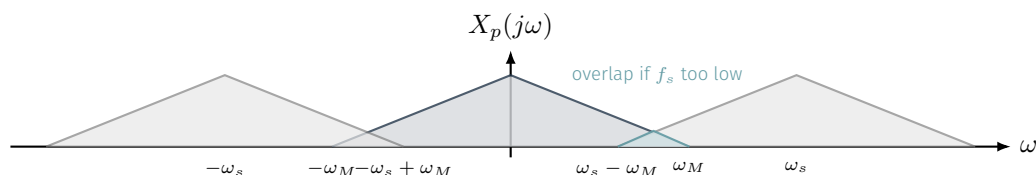


Problem 5.2

Question. Derive the condition $\omega_s > 2\omega_M$ by comparing adjacent spectral copies of a bandlimited spectrum.

Solution. The sampled spectrum contains copies centered at integer multiples of ω_s . No overlap requires

$$\omega_s - \omega_M > \omega_M \implies \omega_s > 2\omega_M.$$



Problem 5.3

Question. A lowpass signal is bandlimited to $|f| \leq 1.6$ kHz. Sketch the sampled spectrum for $f_s = 4.5$ kHz and for $f_s = 2.8$ kHz, and identify which case aliases.

Solution. The Nyquist requirement is

$$f_s > 2f_M = 3.2 \text{ kHz.}$$

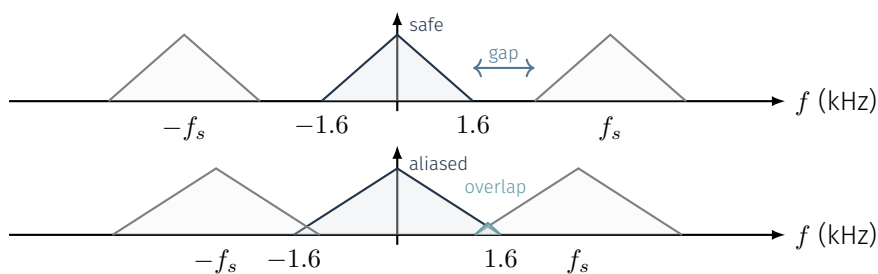
So $f_s = 4.5$ kHz is safe, while $f_s = 2.8$ kHz is too low. For $f_s = 4.5$ kHz, the central copy occupies $[-1.6, 1.6]$ kHz and the nearest shifted copies begin at

$$\pm(4.5 - 1.6) = \pm 2.9 \text{ kHz,}$$

leaving a clear gap. For $f_s = 2.8$ kHz, the adjacent copies begin at

$$\pm(2.8 - 1.6) = \pm 1.2 \text{ kHz,}$$

which lies inside the central support, so overlap occurs.



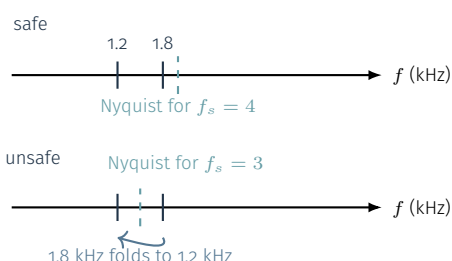
Problem 5.4

Question. A signal contains tones at 1.2 kHz and 1.8 kHz. Give one safe sampling frequency and one unsafe sampling frequency.

Solution. Since the highest tone is $f_M = 1.8$ kHz, a safe choice is $f_s = 4$ kHz and an unsafe one is $f_s = 3$ kHz. With $f_s = 4$ kHz, the Nyquist frequency is 2 kHz, so both tones stay distinct. With $f_s = 3$ kHz, the Nyquist frequency is only 1.5 kHz, so the 1.8 kHz tone folds to

$$f_{\text{alias}} = |1.8 - 3| = 1.2 \text{ kHz},$$

colliding with the 1.2 kHz tone.



5. Ideal Reconstruction, Zero-Order Hold, Linear Interpolation

Problem 6.1

Question. Write the impulse response and frequency response of the ideal reconstruction filter, and decide whether it is causal.

Solution. The sampled spectrum is

$$X_p(j\omega) = \frac{1}{T} \sum_{r=-\infty}^{\infty} X(j(\omega - r\omega_s)).$$

To recover the original signal, the ideal filter must keep only the baseband copy in $|\omega| < \omega_s/2$, and it must multiply by T to cancel the $1/T$ factor in $X_p(j\omega)$. Hence

$$H_{\text{ideal}}(j\omega) = \begin{cases} T, & |\omega| < \omega_s/2, \\ 0, & \text{otherwise,} \end{cases}$$

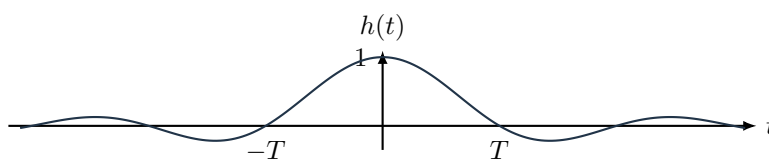
The impulse response is the inverse Fourier transform of this rectangle:

$$h_{\text{ideal}}(t) = \frac{1}{2\pi} \int_{-\omega_s/2}^{\omega_s/2} T e^{j\omega t} d\omega = \frac{T}{2\pi} \left[\frac{e^{j\omega t}}{jt} \right]_{-\omega_s/2}^{\omega_s/2}.$$

Therefore

$$h_{\text{ideal}}(t) = \frac{T}{\pi t} \sin\left(\frac{\omega_s t}{2}\right) = \frac{\sin(\pi t/T)}{\pi t/T} = \text{sinc}\left(\frac{t}{T}\right),$$

because $\omega_s = 2\pi/T$. It is noncausal because the sinc extends over both $t < 0$ and $t > 0$.



Problem 6.2

Question. Derive the impulse response and frequency response of the zero-order hold (ZOH).

Solution. In zero-order hold, one sample $y(nT)$ is held at full weight until the next sample arrives. So the basic reconstruction kernel must be

$$h_0(t) = \begin{cases} 1, & 0 \leq t < T, \\ 0, & \text{otherwise,} \end{cases} = u(t) - u(t - T).$$

Now take its continuous-time Fourier transform:

$$\begin{aligned} H_0(j\omega) &= \int_{-\infty}^{\infty} h_0(t) e^{-j\omega t} dt = \int_0^T e^{-j\omega t} dt \\ &= \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^T = \frac{1 - e^{-j\omega T}}{j\omega}. \end{aligned}$$

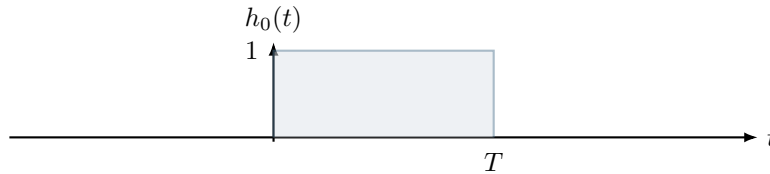
Factor out the midpoint delay:

$$H_0(j\omega) = \frac{e^{-j\omega T/2} (e^{j\omega T/2} - e^{-j\omega T/2})}{j\omega} = e^{-j\omega T/2} \frac{2 \sin(\omega T/2)}{\omega}.$$

Hence

$$H_0(j\omega) = T e^{-j\omega T/2} \operatorname{sinc}\left(\frac{\omega T}{2\pi}\right).$$

ZOH is causal because $h_0(t) = 0$ for $t < 0$.



Problem 6.3

Question. Derive the impulse response and frequency response of linear interpolation.

Solution. For linear interpolation, the weight of $y(nT)$ must rise linearly from 0 to 1 between $(n-1)T$ and nT , then fall linearly from 1 to 0 between nT and $(n+1)T$. So the basic interpolation kernel is

$$h_1(t) = \begin{cases} 1 - |t|/T, & |t| < T, \\ 0, & \text{otherwise,} \end{cases}.$$

This triangle can also be viewed as the overlap of two centered ZOH rectangles:

$$h_1(t) = \frac{1}{T} (h_0(t + T/2) * h_0(t - T/2)).$$

Let $g(t) = h_0(t + T/2)$. Then $g(t)$ is a rectangle centered at the origin, so its transform has no linear-phase term:

$$G(j\omega) = \mathcal{F}\{g(t)\} = T \operatorname{sinc}\left(\frac{\omega T}{2\pi}\right).$$

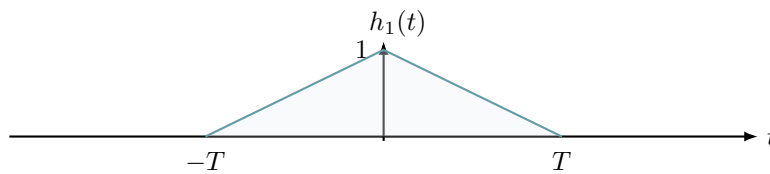
By the convolution property,

$$H_1(j\omega) = \frac{1}{T} (G(j\omega))^2 = \frac{1}{T} \left[T \operatorname{sinc}\left(\frac{\omega T}{2\pi}\right) \right]^2.$$

Therefore

$$H_1(j\omega) = T \operatorname{sinc}^2\left(\frac{\omega T}{2\pi}\right).$$

This symmetric form is noncausal, though a delayed version can be made causal.



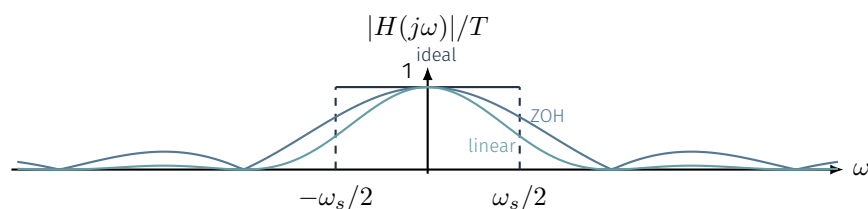
Problem 6.4

Question. Compare the magnitude responses of ideal, ZOH, and linear interpolation at $\omega = \omega_s/4$.

Solution. At $\omega = \omega_s/4 = \pi/(2T)$,

$$\frac{|H_0|}{T} = \operatorname{sinc}\left(\frac{1}{4}\right) = \frac{2\sqrt{2}}{\pi} \approx 0.900, \quad \frac{|H_1|}{T} = \operatorname{sinc}^2\left(\frac{1}{4}\right) \approx 0.811,$$

while the ideal filter has magnitude 1.



Problem 6.5

Question. State one main advantage and one main drawback of each of the three reconstruction methods.

Solution. Ideal reconstruction is perfect in theory but unrealizable in real time. ZOH is extremely easy to implement, but it introduces staircase distortion. Linear interpolation is smoother than ZOH and gives better high-frequency behavior, but it is still not ideal.

ideal	ZOH	linear
perfect reconstruction noncausal	easy hardware droop	smoother than ZOH still approximate

6. Laplace Transform: Definition, Properties, ROC, Poles and Zeros

Problem 7.1

Question. Find the Laplace transform and ROC of $x(t) = (t - 2)e^{-3(t-2)}u(t - 2)$.

Solution.
$$e^{-3t}u(t) \xrightarrow{\mathcal{L}} \frac{1}{s + 3}, \quad \text{ROC : } \text{Re}\{s\} > -3.$$
$$te^{-3t}u(t) \xrightarrow{\mathcal{L}} \frac{1}{(s + 3)^2}, \quad \text{ROC : } \text{Re}\{s\} > -3.$$
Applying the time-shift property to delay the signal by 2,
$$(t - 2)e^{-3(t-2)}u(t - 2) \xrightarrow{\mathcal{L}} e^{-2s} \frac{1}{(s + 3)^2}, \quad \text{ROC : } \text{Re}\{s\} > -3.$$

Problem 7.2

Question. Find the Laplace transform and ROC of $x(t) = t^2e^{-3t}u(t)$.

Solution. Start from
$$e^{-3t}u(t) \xrightarrow{\mathcal{L}} \frac{1}{s + 3}, \quad \text{ROC : } \text{Re}\{s\} > -3.$$
Differentiate once in the s -domain:
$$te^{-3t}u(t) \xrightarrow{\mathcal{L}} -\frac{d}{ds} \left(\frac{1}{s + 3} \right) = \frac{1}{(s + 3)^2}, \quad \text{ROC : } \text{Re}\{s\} > -3.$$
Differentiate once more:
$$t^2e^{-3t}u(t) \xrightarrow{\mathcal{L}} -\frac{d}{ds} \left(\frac{1}{(s + 3)^2} \right) = \frac{2}{(s + 3)^3}, \quad \text{ROC : } \text{Re}\{s\} > -3.$$

Problem 7.3

Question. Find the Laplace transform and ROC of $x(t) = te^{-t}u(t) + e^{2t}u(-t)$.

Solution.
$$e^{-t}u(t) \xrightarrow{\mathcal{L}} \frac{1}{s + 1}, \quad \text{ROC : } \text{Re}\{s\} > -1.$$

Hence

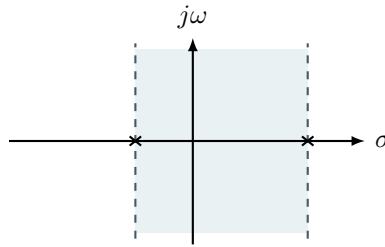
$$te^{-t}u(t) \xrightarrow{\mathcal{L}} -\frac{d}{ds}\left(\frac{1}{s+1}\right) = \frac{1}{(s+1)^2}, \quad \text{ROC} : \text{Re}\{s\} > -1.$$

For the left-sided term,

$$e^{2t}u(-t) \xrightarrow{\mathcal{L}} -\frac{1}{s-2}, \quad \text{ROC} : \text{Re}\{s\} < 2.$$

Therefore

$$X(s) = \frac{1}{(s+1)^2} - \frac{1}{s-2}, \quad \text{ROC} : -1 < \text{Re}\{s\} < 2.$$



Problem 7.4

Question. Find the Laplace transform and ROC of $x(t) = e^{-4(t-1)} \cos(3(t-1))u(t-1)$.

Solution.

$$e^{-4t} \cos(3t)u(t) = \frac{1}{2} \left(e^{(-4+3j)t} + e^{(-4-3j)t} \right) u(t).$$

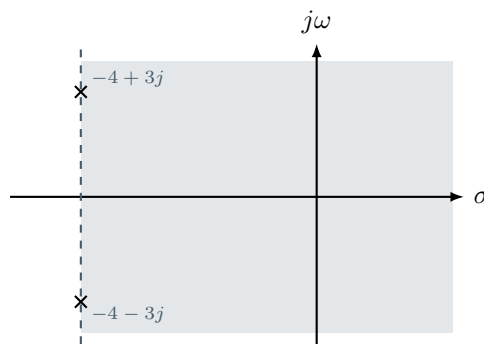
$$e^{(-4+3j)t}u(t) \xrightarrow{\mathcal{L}} \frac{1}{s+4-3j}, \quad e^{(-4-3j)t}u(t) \xrightarrow{\mathcal{L}} \frac{1}{s+4+3j}, \quad \text{ROC} : \text{Re}\{s\} > -4.$$

Therefore

$$e^{-4t} \cos(3t)u(t) \xrightarrow{\mathcal{L}} \frac{1}{2} \left(\frac{1}{s+4-3j} + \frac{1}{s+4+3j} \right) = \frac{s+4}{(s+4)^2 + 9}, \quad \text{ROC} : \text{Re}\{s\} > -4.$$

Applying a delay of 1 multiplies the transform by e^{-s} without changing the ROC:

$$e^{-4(t-1)} \cos(3(t-1))u(t-1) \xrightarrow{\mathcal{L}} e^{-s} \frac{s+4}{(s+4)^2 + 9}, \quad \text{ROC} : \text{Re}\{s\} > -4.$$



Problem 7.5

Question. Find the Laplace transform, ROC, poles, and zeros of $x(t) = te^{-t} \sin(2t)u(t)$.

Solution.

$$e^{-t} \sin(2t)u(t) = \frac{1}{2j} \left(e^{(-1+2j)t} - e^{(-1-2j)t} \right) u(t).$$

$$e^{(-1+2j)t}u(t) \xrightarrow{\mathcal{L}} \frac{1}{s+1-2j}, \quad e^{(-1-2j)t}u(t) \xrightarrow{\mathcal{L}} \frac{1}{s+1+2j}, \quad \text{ROC} : \text{Re}\{s\} > -1.$$

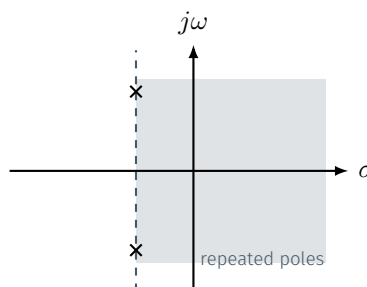
Therefore

$$e^{-t} \sin(2t)u(t) \xrightarrow{\mathcal{L}} \frac{1}{2j} \left(\frac{1}{s+1-2j} - \frac{1}{s+1+2j} \right) = \frac{2}{(s+1)^2 + 4}, \quad \text{ROC} : \text{Re}\{s\} > -1.$$

Differentiate in the s -domain:

$$te^{-t} \sin(2t)u(t) \xrightarrow{\mathcal{L}} -\frac{d}{ds} \left(\frac{2}{(s+1)^2 + 4} \right) = \frac{4(s+1)}{((s+1)^2 + 4)^2}, \quad \text{ROC} : \text{Re}\{s\} > -1.$$

The poles are double poles at $s = -1 \pm 2j$, and the finite zero is at $s = -1$.



Problem 7.6

Question. Find the Laplace transform and ROC of $y(t) = (e^{-t} \sin(3t) u(t)) * (e^{-2t} u(t))$.

Solution.

$$e^{-t} \sin(3t) u(t) = \frac{1}{2j} \left(e^{(-1+3j)t} - e^{(-1-3j)t} \right) u(t).$$

$$e^{(-1+3j)t} u(t) \xrightarrow{\mathcal{L}} \frac{1}{s+1-3j}, \quad e^{(-1-3j)t} u(t) \xrightarrow{\mathcal{L}} \frac{1}{s+1+3j}, \quad \text{ROC: } \text{Re}\{s\} > -1.$$

Hence

$$e^{-t} \sin(3t) u(t) \xrightarrow{\mathcal{L}} \frac{1}{2j} \left(\frac{1}{s+1-3j} - \frac{1}{s+1+3j} \right) = \frac{3}{(s+1)^2 + 9}, \quad \text{ROC: } \text{Re}\{s\} > -1.$$

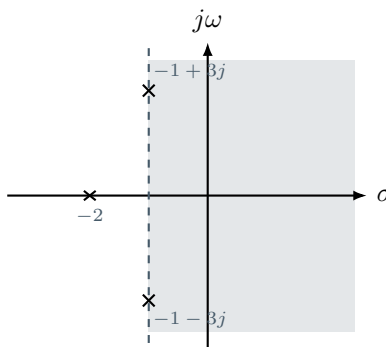
Also

$$e^{-2t} u(t) \xrightarrow{\mathcal{L}} \frac{1}{s+2}, \quad \text{ROC: } \text{Re}\{s\} > -2.$$

By the **convolution property**, convolution in time becomes multiplication in the s -domain, so

$$Y(s) = \frac{3}{((s+1)^2 + 9)(s+2)}, \quad \text{ROC: } \text{Re}\{s\} > -1 \text{ (intersection)}.$$

Poles: $s = -1 \pm 3j$ and $s = -2$. No finite zeros.



7. Inverse Laplace Transform

Residue Method for Partial Fractions

For rational transforms, partial-fraction coefficients can be obtained directly using residue formulas (same idea you saw from Laurent-series coefficients; no extra complex-analysis machinery needed here).

Let

$$F(s) = \frac{P(s)}{Q(s)}.$$

For a **simple pole** at $s = p$:

$$F(s) = \frac{A}{s-p} + \cdots, \quad A = \text{Res } F(s) = \lim_{s \rightarrow p} (s-p)F(s).$$

For a **double pole** at $s = p$:

$$F(s) = \frac{A_2}{(s-p)^2} + \frac{A_1}{s-p} + \cdots, \\ A_2 = \lim_{s \rightarrow p} (s-p)^2 F(s), \quad A_1 = \left. \frac{d}{ds} \left((s-p)^2 F(s) \right) \right|_{s=p}.$$

For a pole of **order** m at $s = p$:

$$F(s) = \frac{A_m}{(s-p)^m} + \frac{A_{m-1}}{(s-p)^{m-1}} + \cdots + \frac{A_1}{s-p} + \cdots, \quad A_k = \frac{1}{(m-k)!} \left. \frac{d^{m-k}}{ds^{m-k}} \left((s-p)^m F(s) \right) \right|_{s=p}.$$

The same formulas work in the z -transform section by setting $r = z^{-1}$, turning factors into $(1 - ar)$.

Problem 8.1

Question. Invert $X(s) = \frac{3s+7}{(s+1)(s+2)}$ with ROC $\text{Re}\{s\} > -1$.

Solution. Partial-fraction expansion:

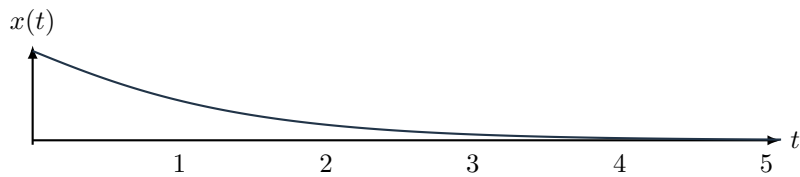
$$\frac{3s+7}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}.$$

Using residues at the simple poles,

$$A = \lim_{s \rightarrow -1} (s+1) \frac{3s+7}{(s+1)(s+2)} = 4, \quad B = \lim_{s \rightarrow -2} (s+2) \frac{3s+7}{(s+1)(s+2)} = -1.$$

Since the ROC is to the right of both poles, both terms are right-sided:

$$x(t) = (4e^{-t} - e^{-2t})u(t).$$

**Problem 8.2**

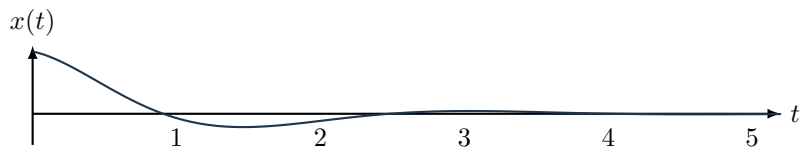
Question. Invert $X(s) = \frac{2s+3}{s^2+2s+5}$ with ROC $\text{Re}\{s\} > -1$.

Solution. Complete the square: $s^2 + 2s + 5 = (s+1)^2 + 4$. Rewrite the numerator:

$$\frac{2s+3}{(s+1)^2+4} = \frac{2(s+1)+1}{(s+1)^2+4} = \frac{2(s+1)}{(s+1)^2+4} + \frac{1}{2} \cdot \frac{2}{(s+1)^2+4}.$$

Using the standard pairs with the s -domain shift property,

$$x(t) = (2e^{-t} \cos 2t + \frac{1}{2}e^{-t} \sin 2t) u(t).$$

**Problem 8.3**

Question. Invert $X(s) = \frac{-5}{(s+1)(s-4)}$ with ROC $-1 < \text{Re}\{s\} < 4$.

Solution. Partial fractions:

$$\frac{-5}{(s+1)(s-4)} = \frac{1}{s+1} - \frac{1}{s-4}.$$

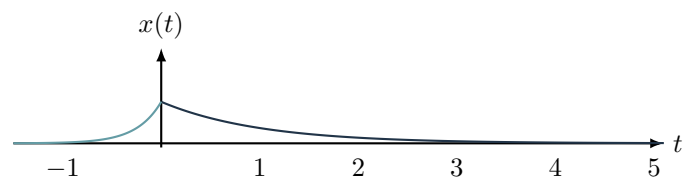
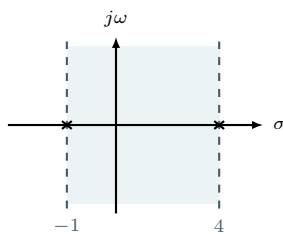
Coefficients from simple-pole residues:

$$\lim_{s \rightarrow -1} (s+1) \frac{-5}{(s+1)(s-4)} = 1, \quad \lim_{s \rightarrow 4} (s-4) \frac{-5}{(s+1)(s-4)} = -1.$$

The ROC is a *strip* between the two poles, so we need a two-sided signal:

- Pole at $s = -1$: to the left of the strip $\Rightarrow$ right-sided term $e^{-t}u(t)$.
- Pole at $s = 4$: to the right of the strip $\Rightarrow$ left-sided term. Since $\frac{1}{s-4} \xrightarrow{\mathcal{L}^{-1}} -e^{4t}u(-t)$ for $\text{Re}\{s\} < 4$, the second term gives $e^{4t}u(-t)$.

$$x(t) = e^{-t}u(t) + e^{4t}u(-t).$$

**Problem 8.4**

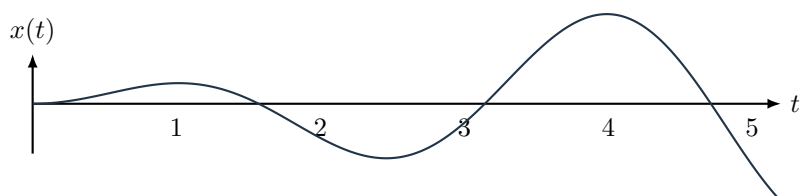
Question. Invert $X(s) = \frac{s}{(s^2+4)^2}$ with ROC $\text{Re}\{s\} > 0$.

Solution.

$$\frac{2}{s^2+4} \xrightarrow{\mathcal{L}^{-1}} \sin(2t)u(t) \quad \Rightarrow \quad \frac{4s}{(s^2+4)^2} \xrightarrow{\mathcal{L}^{-1}} t \sin(2t)u(t).$$

Therefore

$$x(t) = \frac{1}{4} t \sin(2t) u(t).$$



Problem 8.5

Question. Invert $X(s) = \frac{e^{-s}}{s(s+2)}$ with ROC $\text{Re}\{s\} > 0$.

Solution. First invert without the delay:

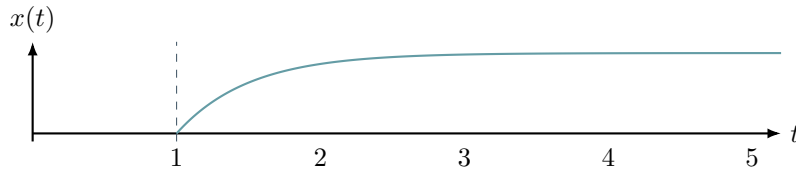
$$\frac{1}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2}, \quad A = \lim_{s \rightarrow 0} s \frac{1}{s(s+2)} = \frac{1}{2}, \quad B = \lim_{s \rightarrow -2} (s+2) \frac{1}{s(s+2)} = -\frac{1}{2}.$$

Hence

$$\frac{1}{s(s+2)} = \frac{1/2}{s} - \frac{1/2}{s+2} \implies y(t) = \frac{1}{2}(1 - e^{-2t})u(t).$$

The factor e^{-s} shifts the result by one unit:

$$x(t) = \frac{1}{2}(1 - e^{-2(t-1)})u(t-1).$$

**Problem 8.6**

Question. Invert $X(s) = \frac{2}{s(s^2+4)}$ with ROC $\text{Re}\{s\} > 0$.

Solution. Write $X(s) = \frac{Y(s)}{s}$ where $Y(s) = \frac{2}{s^2+4}$. Since $Y(s) \xrightarrow{\mathcal{L}^{-1}} \sin(2t)u(t)$, the time-integration property gives

$$x(t) = \int_0^t \sin(2\tau) d\tau \cdot u(t) = \left[-\frac{\cos(2\tau)}{2} \right]_0^t u(t) = \frac{1}{2}(1 - \cos(2t))u(t).$$

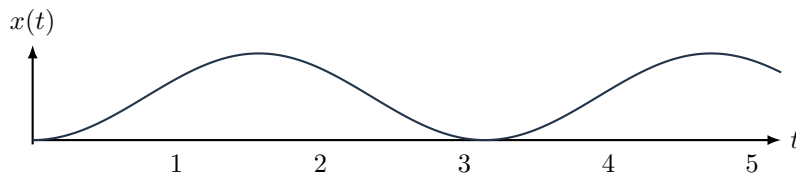
Verification via PFE:

$$\frac{2}{s(s^2+4)} = \frac{A}{s} + \frac{Bs+C}{s^2+4}, \quad A = \lim_{s \rightarrow 0} s \frac{2}{s(s^2+4)} = \frac{1}{2}.$$

Since the numerator of the remaining term is odd/even matched, $C = 0$. Matching s -coefficients gives $B = -\frac{1}{2}$, so

$$\frac{2}{s(s^2+4)} = \frac{1/2}{s} - \frac{s/2}{s^2+4},$$

which leads to the same $x(t)$.

**Problem 8.7**

Question. Invert $X(s) = \frac{s}{s+2}$ with ROC $\text{Re}\{s\} > -2$.

Solution. The fraction is *improper* (numerator degree $\geq$ denominator degree). Perform polynomial long division first:

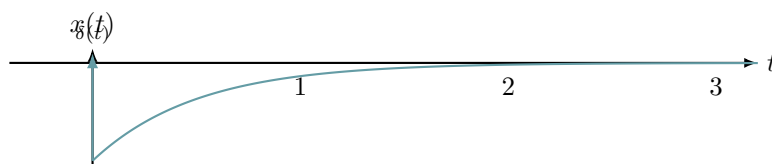
$$\frac{s}{s+2} = 1 - \frac{2}{s+2}.$$

Now $\mathcal{L}^{-1}\{1\} = \delta(t)$ and $\mathcal{L}^{-1}\left\{\frac{2}{s+2}\right\} = 2e^{-2t}u(t)$, so

$$x(t) = \delta(t) - 2e^{-2t}u(t).$$

This is the derivative of $e^{-2t}u(t)$: applying the differentiation property directly,

$$s \cdot \frac{1}{s+2} = s \cdot \mathcal{L}\{e^{-2t}u(t)\} \xrightarrow{\mathcal{L}^{-1}} \frac{d}{dt}[e^{-2t}u(t)] = \delta(t) - 2e^{-2t}u(t). \checkmark$$



8. Applications of Laplace Transform

Problem 9.1

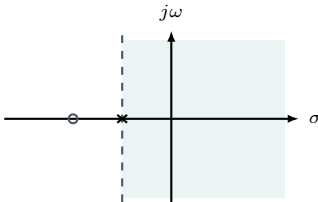
Question. For $H(s) = \frac{s+2}{(s+1)(s+2)}$, determine the admissible ROC choices and the corresponding causality/stability properties.

Solution. After cancellation,
$$H(s) = \frac{1}{s+1}.$$

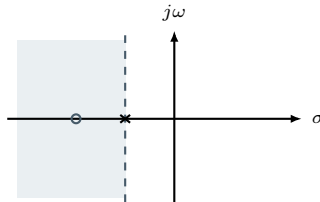
So the only admissible ROCs are

$$\text{Re}\{s\} > -1 \quad \text{or} \quad \text{Re}\{s\} < -1.$$

The right-half ROC gives the causal and stable realization $e^{-t}u(t)$. The left-half ROC gives $-e^{-t}u(-t)$, which is anti-causal and unstable.



$\text{Re}\{s\} > -1$: causal, stable



$\text{Re}\{s\} < -1$: anti-causal, unstable

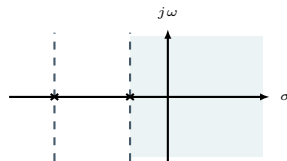
Problem 9.2

Question. For $H(s) = \frac{1}{(s+1)^2(s+3)}$, determine the admissible ROC choices and the corresponding causality/stability properties.

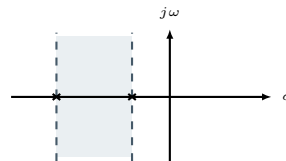
Solution. The poles are at $s = -1$ (double) and $s = -3$, all in the left half-plane. Therefore

$$\text{Re}\{s\} > -1 \Rightarrow \text{causal and stable } (j\omega\text{-axis included}),$$
$$\text{Re}\{s\} < -3 \Rightarrow \text{anti-causal and unstable},$$
$$-3 < \text{Re}\{s\} < -1 \Rightarrow \text{two-sided and unstable } (j\omega\text{-axis not included}).$$

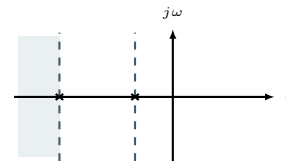
Because all poles lie in the LHP, the causal realization is also stable.



$\text{Re}\{s\} > -1$: causal, stable



$-3 < \text{Re}\{s\} < -1$: two-sided, unstable

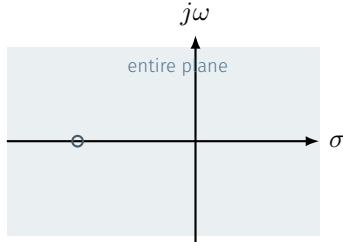


$\text{Re}\{s\} < -3$: anti-causal, unstable

Problem 9.3

Question. Decide whether the Fourier transform exists for $X(s) = \frac{1 - e^{-3(s+2)}}{s+2}$. Interpret the result in time domain.

Solution. This corresponds to $x(t) = e^{-2t}(u(t) - u(t-3))$, which is finite-duration. Its ROC is the entire s -plane, so the $j\omega$ -axis is included and the Fourier transform exists. The apparent pole at -2 is removable.



Problem 9.4

Question. Solve $y'(t) + y(t) = e^{-2(t-1)}u(t-1)$ with $y(0^-) = 0$.

Solution. Using $\mathcal{L}\{y'(t)\} = sY(s)$,

$$sY(s) + Y(s) = \frac{e^{-s}}{s+2} \implies (s+1)Y(s) = \frac{e^{-s}}{s+2}.$$

Hence

$$Y(s) = \frac{e^{-s}}{(s+1)(s+2)}.$$

Since

$$\frac{1}{(s+1)(s+2)} = \frac{1}{s+1} - \frac{1}{s+2},$$

the poles are at -1 and -2 , so the admissible ROCs are

$$\operatorname{Re}\{s\} > -1, \quad -2 < \operatorname{Re}\{s\} < -1, \quad \operatorname{Re}\{s\} < -2.$$

These give the three inverse candidates

$$\operatorname{Re}\{s\} > -1 \implies y_1(t) = \left(e^{-(t-1)} - e^{-2(t-1)}\right)u(t-1),$$

$$-2 < \operatorname{Re}\{s\} < -1 \implies y_2(t) = -e^{-(t-1)}u(1-t) - e^{-2(t-1)}u(t-1),$$

$$\operatorname{Re}\{s\} < -2 \implies y_3(t) = \left(e^{-2(t-1)} - e^{-(t-1)}\right)u(1-t).$$

Now apply the initial condition:

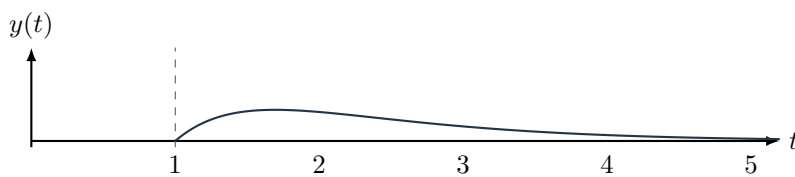
$$y(0^-) = 0.$$

Only $y_1(t)$ satisfies $y(0^-) = 0$, so the correct ROC is

$$\text{ROC} : \operatorname{Re}\{s\} > -1,$$

and the solution is

$$y(t) = \left(e^{-(t-1)} - e^{-2(t-1)}\right)u(t-1).$$



Problem 9.5

Question. Solve $y''(t) + 3y'(t) + 2y(t) = e^{-(t-1)}u(t-1)$ with zero initial conditions.

Solution. Using $\mathcal{L}\{y'(t)\} = sY(s)$ and $\mathcal{L}\{y''(t)\} = s^2Y(s)$,

$$s^2Y(s) + 3sY(s) + 2Y(s) = \frac{e^{-s}}{s+1} \implies (s^2 + 3s + 2)Y(s) = \frac{e^{-s}}{s+1}.$$

Therefore

$$Y(s) = \frac{e^{-s}}{(s+1)^2(s+2)}.$$

For

$$F(s) = \frac{1}{(s+1)^2(s+2)} = \frac{A}{s+1} + \frac{B}{(s+1)^2} + \frac{C}{s+2},$$

residues give

$$C = \lim_{s \rightarrow -2} (s+2)F(s) = 1, \quad B = \lim_{s \rightarrow -1} (s+1)^2 F(s) = 1,$$

and for the simple-over-repeated coefficient,

$$A = \frac{d}{ds} \left((s+1)^2 F(s) \right) \Big|_{s=-1} = -1.$$

Therefore

$$\frac{1}{(s+1)^2(s+2)} = \frac{1}{s+2} - \frac{1}{s+1} + \frac{1}{(s+1)^2},$$

so the admissible ROCs are

$$\operatorname{Re}\{s\} > -1, \quad -2 < \operatorname{Re}\{s\} < -1, \quad \operatorname{Re}\{s\} < -2.$$

The corresponding inverses are

$$\operatorname{Re}\{s\} > -1 \implies y_1(t) = \left(e^{-2(t-1)} - e^{-(t-1)} + (t-1)e^{-(t-1)}\right)u(t-1),$$

$$-2 < \operatorname{Re}\{s\} < -1 \implies y_2(t) = e^{-2(t-1)}u(t-1) + (2-t)e^{-(t-1)}u(1-t),$$

$$\operatorname{Re}\{s\} < -2 \implies y_3(t) = \left(-e^{-2(t-1)} + (2-t)e^{-(t-1)}\right)u(1-t).$$

Now use the initial conditions:

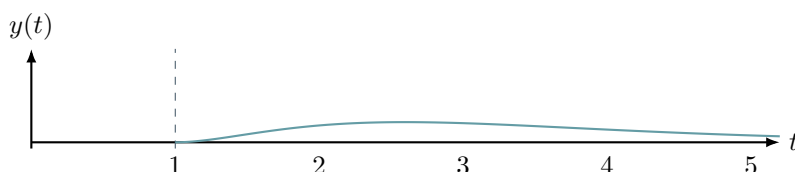
$$y(0^-) = 0, \quad y'(0^-) = 0, \quad y''(0^-) = 0.$$

Therefore the only admissible choice is

$$\text{ROC} : \operatorname{Re}\{s\} > -1.$$

For this ROC the signal is zero for $t < 1$, so $y'(0^-) = 0$ also holds. Hence

$$y(t) = \left(e^{-2(t-1)} - e^{-(t-1)} + (t-1)e^{-(t-1)}\right)u(t-1).$$



Problem 9.6

Question. A causal LTI system is described by the LCCDE

$$\frac{d^2 y}{dt^2} + 5 \frac{dy}{dt} + 6y(t) = \frac{dx}{dt} + 3x(t).$$

Find $H(s)$, sketch the pole-zero plot, determine whether the system is stable, and find the impulse response $h(t)$.

Solution. Transform the LCCDE directly:

$$s^2 Y(s) + 5sY(s) + 6Y(s) = sX(s) + 3X(s).$$

Hence

$$H(s) = \frac{Y(s)}{X(s)} = \frac{s+3}{s^2+5s+6} = \frac{s+3}{(s+2)(s+3)} = \frac{1}{s+2}.$$

There is a pole-zero cancellation at $s = -3$. The remaining pole is at $s = -2$ and there are no finite zeros. The algebraic form $H(s) = 1/(s+2)$ admits two possible ROCs:

$$\text{Re}\{s\} > -2 \quad \text{or} \quad \text{Re}\{s\} < -2.$$

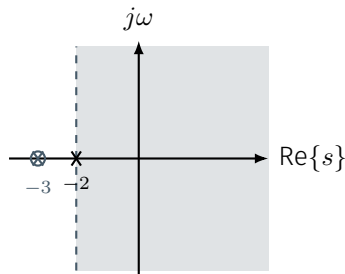
Since the system is stated to be causal, we choose

$$\text{ROC} : \text{Re}\{s\} > -2.$$

Hence the impulse response is

$$h(t) = e^{-2t} u(t).$$

Because this ROC includes the $j\omega$ -axis, the system is **stable**.



9. z-Transform: Definition, Properties, ROC, Poles and Zeros

Problem 10.1

Question. Find the z-transform, poles, zeros, and ROC of $x[n] = \left(\frac{3}{4}\right)^n \cos\left(\frac{\pi n}{3}\right) u[n]$.

Solution. Using Euler's formula,

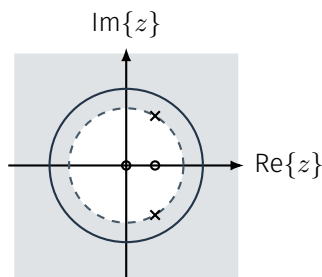
$$x[n] = \frac{1}{2} \left(\frac{3}{4} e^{j\pi/3}\right)^n u[n] + \frac{1}{2} \left(\frac{3}{4} e^{-j\pi/3}\right)^n u[n].$$

Therefore

$$X(z) = \frac{1 - \frac{3}{8} z^{-1}}{1 - \frac{3}{4} z^{-1} + \frac{9}{16} z^{-2}}.$$

Poles: $z = \frac{3}{4} e^{\pm j\pi/3}$. Zeros: $z = 0, \frac{3}{8}$. ROC:

$$|z| > \frac{3}{4}.$$

**Problem 10.2**

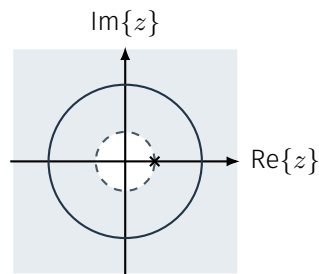
Question. Find the z-transform and ROC of $x[n] = (n-2) \left(\frac{1}{3}\right)^{n-2} u[n-2]$.

Solution. Let $x_0[n] = n \left(\frac{1}{3}\right)^n u[n]$. Then

$$n \left(\frac{1}{3}\right)^n u[n] \xrightarrow{\mathcal{Z}} \frac{\frac{1}{3} z^{-1}}{\left(1 - \frac{1}{3} z^{-1}\right)^2}, \quad X_0(z) = \frac{\frac{1}{3} z^{-1}}{\left(1 - \frac{1}{3} z^{-1}\right)^2}.$$

Since $x[n] = x_0[n - 2]$,

$$X(z) = z^{-2} \frac{\frac{1}{3}z^{-1}}{\left(1 - \frac{1}{3}z^{-1}\right)^2}, \quad \text{ROC: } |z| > \frac{1}{3}.$$



Problem 10.3

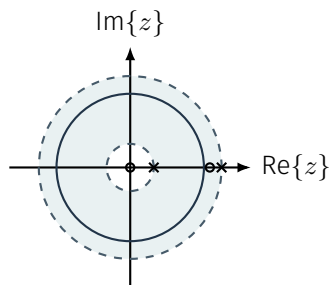
Question. Find the z-transform, poles, zeros, and ROC of $x[n] = \left(\frac{1}{4}\right)^n u[n] - 2^n u[-n - 1]$.

Solution.

$$X(z) = \frac{1}{1 - \frac{1}{4}z^{-1}} + \frac{1}{1 - 2z^{-1}} = \frac{2\left(1 - \frac{9}{8}z^{-1}\right)}{\left(1 - \frac{1}{4}z^{-1}\right)\left(1 - 2z^{-1}\right)}.$$

Poles: $z = \frac{1}{4}, 2$. Zeros: $z = 0, \frac{9}{8}$. ROC:

$$\frac{1}{4} < |z| < 2.$$



Problem 10.4

Question. Find the z-transform and zero pattern of $x[n] = \delta[n] - 2\delta[n - 1] + 2\delta[n - 2] - \delta[n - 3]$.

Solution.

$$X(z) = 1 - 2z^{-1} + 2z^{-2} - z^{-3}.$$

Multiplying by z^3 ,

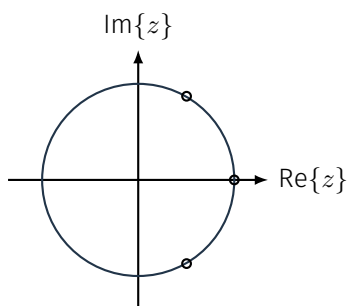
$$z^3 X(z) = z^3 - 2z^2 + 2z - 1 = (z - 1)(z^2 - z + 1).$$

So the zeros are

$$z = 1, \quad z = e^{j\pi/3}, \quad z = e^{-j\pi/3},$$

and because the sequence is finite and right-sided,

$$\text{ROC: } z \neq 0.$$



Problem 10.5

Question. Find the z-transform, poles, zeros, and ROC of $x[n] = \left(\frac{1}{2}\right)^{n-1} \cos\left(\frac{\pi(n-1)}{3}\right) u[n - 1]$.

Solution. For the undelayed sequence,

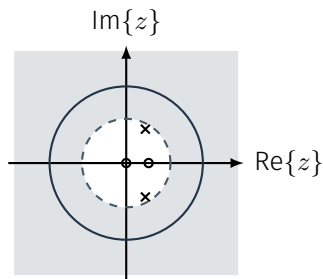
$$X_0(z) = \frac{1 - \frac{1}{4}z^{-1}}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}.$$

A unit delay gives

$$X(z) = z^{-1} X_0(z) = \frac{z^{-1} \left(1 - \frac{1}{4}z^{-1}\right)}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}.$$

The poles are $z = \frac{1}{2}e^{\pm j\pi/3}$, the zeros are $z = 0, \frac{1}{4}$, and

$$\text{ROC} : |z| > \frac{1}{2}.$$



Problem 10.6

Question. Find the z-transform, poles, zeros, and ROC of $y[n] = \left(\left(\frac{1}{2}\right)^n u[n]\right) * \left(\left(\frac{1}{3}\right)^n u[n]\right)$.

Solution. By the **convolution property**, convolution in time becomes multiplication in the z -domain:

$$\left(\frac{1}{2}\right)^n u[n] \xrightarrow{\mathcal{Z}} \frac{1}{1 - \frac{1}{2}z^{-1}}, \quad \text{ROC} : |z| > \frac{1}{2},$$

$$\left(\frac{1}{3}\right)^n u[n] \xrightarrow{\mathcal{Z}} \frac{1}{1 - \frac{1}{3}z^{-1}}, \quad \text{ROC} : |z| > \frac{1}{3}.$$

Therefore

$$Y(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})}, \quad \text{ROC} : |z| > \frac{1}{2} \text{ (intersection)}.$$

Poles: $z = \frac{1}{2}, \frac{1}{3}$. No finite zeros (numerator is 1).

Verification via partial fractions:

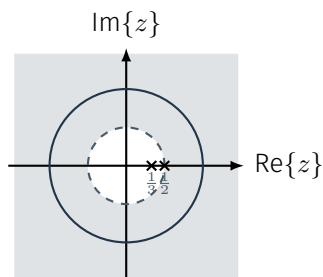
$$Y(z) = \frac{A}{1 - \frac{1}{2}z^{-1}} + \frac{B}{1 - \frac{1}{3}z^{-1}}.$$

Let $r = z^{-1}$. Then

$$A = (1 - \frac{1}{2}r) \frac{1}{(1 - \frac{1}{2}r)(1 - \frac{1}{3}r)} \Big|_{r=2} = 3, \quad B = (1 - \frac{1}{3}r) \frac{1}{(1 - \frac{1}{2}r)(1 - \frac{1}{3}r)} \Big|_{r=3} = -2.$$

So

$$Y(z) = \frac{3}{1 - \frac{1}{2}z^{-1}} - \frac{2}{1 - \frac{1}{3}z^{-1}} \implies y[n] = 3\left(\frac{1}{2}\right)^n u[n] - 2\left(\frac{1}{3}\right)^n u[n].$$



Problem 10.7

Question. Let $x[n] = \left(\frac{1}{2}\right)^n u[n]$. Find the z-transform and ROC of the running sum $y[n] = \sum_{k=-\infty}^n x[k]$.

Solution. By the **accumulation property**,

$$Y(z) = \frac{X(z)}{1 - z^{-1}} = \frac{1}{(1 - z^{-1})(1 - \frac{1}{2}z^{-1})}, \quad \text{ROC} : |z| > 1.$$

The new pole at $z = 1$ from the accumulator restricts the ROC to $|z| > 1$.

Verification:

$$Y(z) = \frac{A}{1 - z^{-1}} + \frac{B}{1 - \frac{1}{2}z^{-1}}.$$

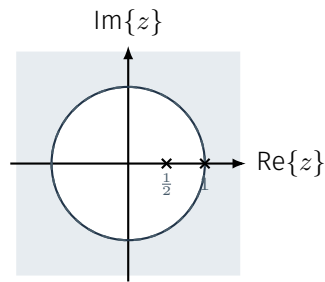
With $r = z^{-1}$,

$$A = (1 - r) \frac{1}{(1 - r)(1 - \frac{1}{2}r)} \Big|_{r=1} = 2, \quad B = (1 - \frac{1}{2}r) \frac{1}{(1 - r)(1 - \frac{1}{2}r)} \Big|_{r=2} = -1.$$

Hence

$$Y(z) = \frac{2}{1 - z^{-1}} - \frac{1}{1 - \frac{1}{2}z^{-1}},$$

so $y[n] = \left(2 - \left(\frac{1}{2}\right)^n\right)u[n]$. Check: $y[0] = 1$, $y[1] = \frac{3}{2}$, $y[\infty] \rightarrow 2$.



10. Inverse z-Transform

Problem 11.1

Question. Invert $X(z) = \frac{2 - z^{-1}}{(1 - \frac{1}{4}z^{-1})(1 - \frac{3}{4}z^{-1})}$ for $|z| > \frac{3}{4}$.

Solution. Partial fractions:

$$X(z) = \frac{A}{1 - \frac{1}{4}z^{-1}} + \frac{B}{1 - \frac{3}{4}z^{-1}}.$$

Let $r = z^{-1}$. Then

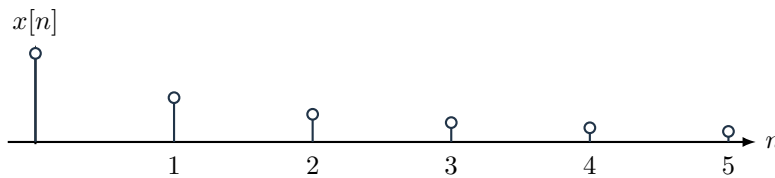
$$A = (1 - \frac{1}{4}r) \frac{2 - r}{(1 - \frac{1}{4}r)(1 - \frac{3}{4}r)} \Big|_{r=\frac{4}{3}} = 1, \quad B = (1 - \frac{3}{4}r) \frac{2 - r}{(1 - \frac{1}{4}r)(1 - \frac{3}{4}r)} \Big|_{r=\frac{4}{3}} = 1.$$

Therefore

$$\frac{2 - z^{-1}}{(1 - \frac{1}{4}z^{-1})(1 - \frac{3}{4}z^{-1})} = \frac{1}{1 - \frac{1}{4}z^{-1}} + \frac{1}{1 - \frac{3}{4}z^{-1}}.$$

Since $|z| > 3/4$, both terms are right-sided:

$$x[n] = \left(\frac{1}{4}\right)^n u[n] + \left(\frac{3}{4}\right)^n u[n].$$



Problem 11.2

Question. Invert $X(z) = \frac{1}{(1 - 2z^{-1})(1 - \frac{1}{3}z^{-1})}$ for $\frac{1}{3} < |z| < 2$.

Solution. Partial fractions:

$$X(z) = \frac{A}{1 - 2z^{-1}} + \frac{B}{1 - \frac{1}{3}z^{-1}}.$$

Let $r = z^{-1}$. Then

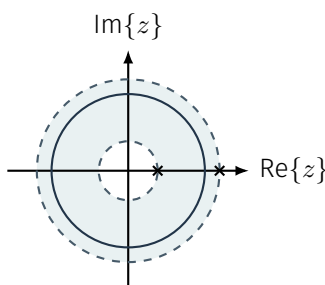
$$A = (1 - 2r) \frac{1}{(1 - 2r)(1 - \frac{1}{3}r)} \Big|_{r=\frac{1}{2}} = \frac{6}{5}, \quad B = (1 - \frac{1}{3}r) \frac{1}{(1 - 2r)(1 - \frac{1}{3}r)} \Big|_{r=3} = -\frac{1}{5}.$$

So

$$\frac{1}{(1 - 2z^{-1})(1 - \frac{1}{3}z^{-1})} = \frac{6/5}{1 - 2z^{-1}} - \frac{1/5}{1 - \frac{1}{3}z^{-1}}.$$

With $\frac{1}{3} < |z| < 2$, the pole at $z = 2$ gives a left-sided term and the pole at $z = \frac{1}{3}$ gives a right-sided term:

$$x[n] = -\frac{6}{5} \cdot 2^n u[-n - 1] - \frac{1}{5} \left(\frac{1}{3}\right)^n u[n].$$



Problem 11.3

Question. Invert $X(z) = \frac{1 - \frac{5}{6}z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})}$ for $|z| < \frac{1}{3}$.

Solution. Partial fractions:

$$X(z) = \frac{A}{1 - \frac{1}{2}z^{-1}} + \frac{B}{1 - \frac{1}{3}z^{-1}}.$$

Let $r = z^{-1}$. Then

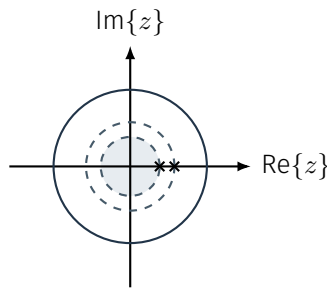
$$A = (1 - \frac{1}{2}r) \frac{1 - \frac{5}{6}r}{(1 - \frac{1}{2}r)(1 - \frac{1}{3}r)} \Big|_{r=2} = -2, \quad B = (1 - \frac{1}{3}r) \frac{1 - \frac{5}{6}r}{(1 - \frac{1}{2}r)(1 - \frac{1}{3}r)} \Big|_{r=3} = 3.$$

Therefore

$$\frac{1 - \frac{5}{6}z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})} = \frac{-2}{1 - \frac{1}{2}z^{-1}} + \frac{3}{1 - \frac{1}{3}z^{-1}}.$$

With $|z| < \frac{1}{3}$, both poles are outside the ROC, so both terms are left-sided:

$$x[n] = 2 \left(\frac{1}{2}\right)^n u[-n-1] - 3 \left(\frac{1}{3}\right)^n u[-n-1].$$

**Problem 11.4**

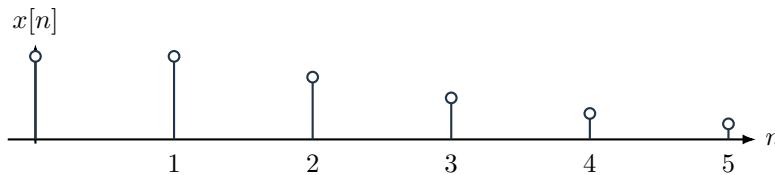
Question. Invert $X(z) = \frac{1 + \frac{1}{2}z^{-1}}{(1 - \frac{1}{2}z^{-1})^2}$ for $|z| > \frac{1}{2}$.

Solution. Write

$$\frac{1 + \frac{1}{2}z^{-1}}{(1 - \frac{1}{2}z^{-1})^2} = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{\frac{1}{2}z^{-1}}{(1 - \frac{1}{2}z^{-1})^2}.$$

Therefore

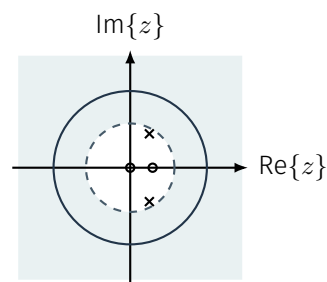
$$x[n] = \left(\frac{1}{2}\right)^n u[n] + n \left(\frac{1}{2}\right)^n u[n] = (n+1) \left(\frac{1}{2}\right)^n u[n].$$

**Problem 11.5**

Question. Invert $X(z) = z^{-1} \frac{1 - \frac{1}{4}z^{-1}}{1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}}$ for $|z| > \frac{1}{2}$.

Solution. This is the delayed damped-cosine pair:

$$x[n] = \left(\frac{1}{2}\right)^{n-1} \cos\left(\frac{\pi(n-1)}{3}\right) u[n-1].$$



11. Applications of z-Transform: Causality, Stability from ROC

Problem 12.1

Question. For $H(z) = \frac{1 - 2z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})}$, determine the valid ROC choices and the corresponding causality/stability/DTFT conclusions.

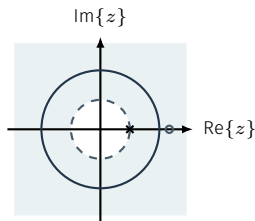
Solution. The factor $(1 - 2z^{-1})$ cancels, so

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}.$$

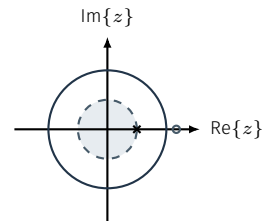
The valid ROCs are only $|z| > \frac{1}{2}$ and $|z| < \frac{1}{2}$. Therefore

$$|z| > \frac{1}{2} \Rightarrow \text{causal, stable, DTFT exists,}$$

$$|z| < \frac{1}{2} \Rightarrow \text{anti-causal, unstable, no DTFT.}$$



$|z| > \frac{1}{2}$: causal, stable, DTFT exists



$|z| < \frac{1}{2}$: anti-causal, unstable, no DTFT

Problem 12.2

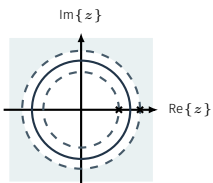
Question. For $H(z) = \frac{1}{(1 - 0.8z^{-1})(1 - 1.25z^{-1})}$, determine whether any realization can be both stable and causal.

Solution. The pole radii are 0.8 and 1.25. Thus

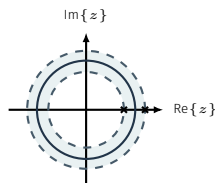
$$|z| > 1.25 \Rightarrow \text{causal but unstable,} \quad |z| < 0.8 \Rightarrow \text{anti-causal and unstable,}$$

$$0.8 < |z| < 1.25 \Rightarrow \text{stable two-sided.}$$

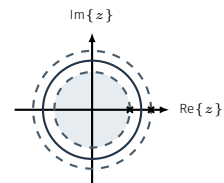
So a stable realization exists, but it cannot be causal.



$|z| > 1.25$: causal, unstable



$0.8 < |z| < 1.25$: stable, two-sided



$|z| < 0.8$: anti-causal, unstable

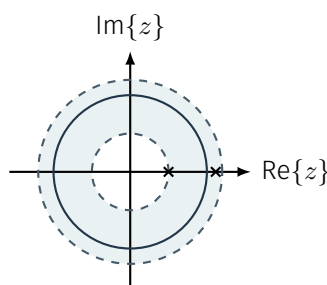
Problem 12.3

Question. Determine whether the DTFT exists for $x[n] = (\frac{1}{2})^n u[n] - 1.2^n u[-n - 1]$.

Solution. The ROC is

$$0.5 < |z| < 1.2,$$

which includes the unit circle. Hence the DTFT exists. Because the sequence is two-sided, it is not causal.



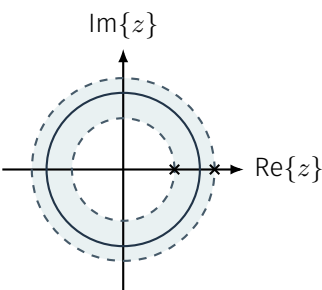
Problem 12.4

Question. Given $h[n] = (\frac{2}{3})^n u[n] - (\frac{4}{3})^n u[-n - 1]$, find $H(z)$ and its ROC, and determine whether the system is causal, stable, and whether the DTFT of $h[n]$ exists.

Solution.

$$H(z) = \frac{1}{1 - \frac{2}{3}z^{-1}} + \frac{1}{1 - \frac{4}{3}z^{-1}} = \frac{2 - 2z^{-1}}{(1 - \frac{2}{3}z^{-1})(1 - \frac{4}{3}z^{-1})}, \quad \text{ROC: } \frac{2}{3} < |z| < \frac{4}{3}.$$

Since the ROC does not include $z = \infty$, the system is **not causal**. The unit circle lies inside the annulus ($\frac{2}{3} < 1 < \frac{4}{3}$), so the system is **stable** and the DTFT exists.



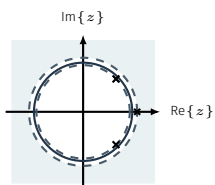
Problem 12.5

Question. A system has poles at $0.95e^{\pm j\pi/4}$ and 1.1. Determine whether there exists a stable realization, a causal realization, and a realization that is both.

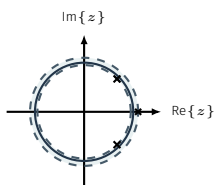
Solution. A causal realization always uses $|z| > 1.1$, so it exists but is unstable. A stable realization also exists because

$$0.95 < |z| < 1.1$$

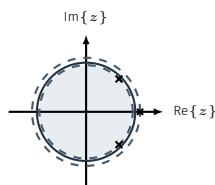
contains the unit circle. However, that stable realization is two-sided, so no realization is both stable and causal.



$|z| > 1.1$: causal, unstable



$0.95 < |z| < 1.1$: stable, two-sided



$|z| < 0.95$: anti-causal, unstable